

CrossMark
click for updatesCite this: *RSC Adv.*, 2016, 6, 95663

Nanostructured layered Sn_3O_4 for hydrogen production and dye degradation under sunlight

S. D. Balgude,^a Y. A. Sethi,^a B. B. Kale,^{*a} N. R. Munirathnam,^a D. P. Amalnerkar^b and P. V. Adhyapak^{*a}

Nanostructured layered Sn_3O_4 nanoplatelets were synthesized *via* a facile hydrothermal method. At low pH, the formation of SnO_2 was observed due to the existence of Sn(II) . By tuning the pH of the hydrothermal reaction, Sn_3O_4 was conferred. The broadening of XRD peaks reveals the existence of nanocrystalline triclinic Sn_3O_4 , which was also confirmed by Raman spectroscopy. Morphological examination displays the nanostructured layered plate-like structure with diameter of 20 nm and length of 100–150 nm. The optical band gap was observed to be 2.76 eV for Sn_3O_4 . Considering the band gap was in the visible light region, investigations on the photocatalytic production of hydrogen and dye degradation were performed under sunlight. A maximum hydrogen evolution ($3916 \mu\text{mol h}^{-1}/0.5 \text{ g}$) was obtained using the Sn_3O_4 nanoparticles, which is much higher than that found with SnO_2 ($729 \mu\text{mol h}^{-1}/0.5 \text{ g}$) due to narrowing of the band gap. Hydrogen production from H_2S using Sn_3O_4 was reported for the first time. Sn_3O_4 also demonstrates excellent dye degradation activity. The high photocatalytic activity of Sn_3O_4 was attributed to its nanocrystalline nature and narrowing of the band gap.

Received 8th August 2016
Accepted 19th September 2016

DOI: 10.1039/c6ra20058c

www.rsc.org/advances

1 Introduction

Various semiconducting materials such as TiO_2 , ZnO , Fe_2O_3 , SnO_2 , CdS and ZnS have been customarily used as photocatalysts that act as sensitizers for light induced redox-processes. Among such metal oxide semiconductor materials, tin oxides with different forms *viz.* SnO , SnO_2 and Sn_3O_4 , possess a special place owing to their wide-spread applications in catalysis, sensors, transparent electrodes, lithium-ion batteries, solar cells *etc.*^{1–4} Their abundant availability and non-toxicity have made tin oxides popular contenders. Stannous (SnO) and stannic (SnO_2) are two univalent forms of tin oxide with the Sn^{2+} and Sn^{4+} oxidation states. Both compounds exhibit quite different physico-chemical characteristics and have different applications.^{5–7} Similarly, heterovalent tin oxides such as Sn_5O_6 , Sn_3O_4 and Sn_2O_3 have also been reported by some authors both theoretically^{8,9} and experimentally.^{10,11} However, these heterovalent oxide materials have not yet been fully explored due to the difficulties associated with obtaining stable and pure crystalline phases when compared with the stoichiometric monovalent oxides. Sn_3O_4 was reported as the first identified non-stoichiometric tin oxide and has been investigated in terms of its crystal structure later.¹⁰ The Sn_3O_4 band structure

shows an energy gap in the visible light region, which is an added advantage for optical applications.^{12,13} Sn_3O_4 with hierarchical morphologies (*viz.* nanobelt, nanosphere and nanosheet morphologies) has also been successfully synthesized.^{14–17} To the best of our knowledge, there are very few reports on the synthesis of high quality pure Sn_3O_4 using a simple method and its effective utilization as a photocatalyst. C. J. Damaschio *et al.* have reported the growth of Sn_3O_4 nanobelt using a carbothermal reduction process.¹⁵ W. Xu *et al.* have synthesized hierarchical Sn_3O_4 nanoflowers assembled from nanosheets.¹⁸ M. Li *et al.* have shown the effect of pH on the morphology and optical properties of Sn_3O_4 .¹⁹ P. H. Suman *et al.* have studied the gas sensing properties of SnO_2 , SnO and Sn_3O_4 nanobelts.²⁰ These reports have enlightened the scope to study such materials in view of their photocatalytic ability. In the context of Sn_3O_4 as a photocatalyst, very limited studies have been reported. In view of this, we have focused on the synthesis of Sn_3O_4 and its photocatalytic activity.

We report the simple hydrothermal synthesis of nanostructured layered Sn_3O_4 . The as-synthesized samples were characterized using XRD, FE-SEM, FE-TEM, Raman, PL and UV-visible spectroscopy. The photocatalytic activities of the as-synthesized samples were evaluated by hydrogen evolution *via* H_2S splitting under sunlight irradiation as well as the degradation of methylene blue (MB) dye. To the best of our knowledge, this is the first report on hydrogen generation *via* H_2S splitting using a Sn_3O_4 photocatalyst.

^aCentre for Materials for Electronics Technology, Panchawati, Pashan Road, Pune 411008, India. E-mail: bbkale@cmet.gov.in; adhyapak@cmet.gov.in

^bSchool of Mechanical Engineering, Sungkyunkwan University, 300, Cheoncheon-dong, Jangnan-gu, Suwon, Gyeonggi, South Korea

2 Experimental

Materials

All reagents used were of AR grade and used as received without further purification. Stannous chloride dihydrate ($\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$) and sodium hydroxide (NaOH) were purchased from Merck (India). Methylene Blue (MB) was purchased from SD Fine Chemical Co. Double distilled water was used as the solvent for the synthesis.

Synthesis of Sn_3O_4 and SnO_2 nanoparticles

For the preparation of Sn_3O_4 , 7 g of $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ was dissolved in 120 mL of water using a magnetic stirrer at 500 rpm. A 0.1 M solution of NaOH was added dropwise to this solution till the pH of the solution becomes highly alkaline (pH 13). The reaction mixture appears as a milky white colloidal suspension at this stage. The resultant mixture was magnetically stirred for 30 min at room temperature (RT) and then sonicated for 15 min using an ultrasonic bath. The solution was then transferred into a 250 mL Teflon-lined stainless-steel autoclave. The autoclave was sealed and placed in a pre-heated oven at 180 °C for 16 h and subsequently allowed to cool down to RT naturally. The obtained precipitate was centrifuged, washed several times with distilled water and absolute ethanol, and dried at 60 °C for 12 h (sample A). Another set of experiments was also carried using the same process with the exception of adjusting the pH to 5 and the hydrothermal treatment at 120 °C for 16 h (sample B). The resultant powders were stored in a desiccator.

Physicochemical characterization

Identification of the phases of the obtained samples was carried out by X-ray diffraction analysis (XRD) (model Rigaku Miniflex X-ray diffractometer) with $\text{Cu K}\alpha$ irradiation at $\lambda = 1.5406 \text{ \AA}$. The surface morphology of the as-prepared samples was investigated by field emission scanning electron microscopy (FESEM) using a JEOL-JSM Model 6700F. Field emission transmission electron microscopy (FETEM) images of the Sn_3O_4 samples was performed on a JEOL, JEM-2200 FS transmission electron microscope equipped with field emission gun. Raman spectra were recorded on a Jobin Yvon T64000 triple grating spectrometer equipped with a liquid nitrogen cooled charge coupled device. Optical absorption spectra were generated using a JASCOV-570 spectrophotometer. Room temperature photoluminescence spectra (PL) of the aqueous suspensions were recorded on an F-2500 fluorescence spectrophotometer. For the PL measurements, the water suspensions of the samples were excited at a wavelength of 300 nm.

Photocatalytic study for the degradation of methylene blue (MB) dye

Photocatalytic measurements of the synthesized product were performed during the degradation of MB. P25 was used as a standard material for comparison. All photocatalytic experiments were carried out in a Pyrex beaker. 50 mg of catalyst was added to 100 mL of a 10 ppm solution of MB and the mixture

was magnetically stirred in the dark for 1 h to achieve an adsorption equilibrium of MB with the catalyst and then exposed to sunlight. The reaction mixture was collected at stipulated time intervals and centrifuged to remove the catalyst for spectrophotometric analysis. The change in MB concentration was monitored by the change in the main λ_{max} centered at 663 nm.

Photocatalytic H_2S splitting for H_2 generation

A cylindrical quartz photochemical thermostatic reactor was filled with 700 mL of 0.5 M aqueous KOH and purged with Ar for 1 h. H_2S was bubbled through the solution at a rate of 2.5 mL min^{-1} at 298 K. H_2S was continuously fed into the system during the photo-reduction. 0.5 g of the as-prepared sample was introduced into the reactor and irradiated with normal solar light with constant stirring. The excess H_2S was trapped in the NaOH solution and the amount of hydrogen evolved was measured using a graduated glass burette.

3 Results and discussion

Fig. 1 depicts the powdered XRD patterns of the as-synthesized Sn_3O_4 and SnO_2 samples at various pH values. The XRD pattern of Sn_3O_4 (sample A) at pH 13 is in good agreement with JCPDS card No. 16-0737.²¹ The pattern shows distinctly broadened peaks at $2\theta = 24.06, 26.92, 31.40, 33.86, 37.13, 49.98$ and 51.79° , which corresponded to the lattice planes (101), (111), (-210), (-122), (130), (-311) and (-132), respectively. The determined lattice constants a , b and c are 4.85, 5.87 and 8.20 Å respectively, which match well with triclinic phase of Sn_3O_4 .²² The XRD spectrum of SnO_2 (sample B) at pH 5 matches with the tetragonal rutile structure of SnO_2 and was in accordance with JCPDS card No. 41-1445. Both the samples exhibit well defined crystalline peaks with no other peaks due to impurity phases.

Further structural and impurity detection investigations were conducted for the as-prepared samples using Raman spectroscopy. Fig. 2 shows the Raman spectra of the as-prepared Sn_3O_4 and SnO_2 samples, which revealed two strong and sharp fundamental peaks located at 138 and 164 cm^{-1} , corresponding

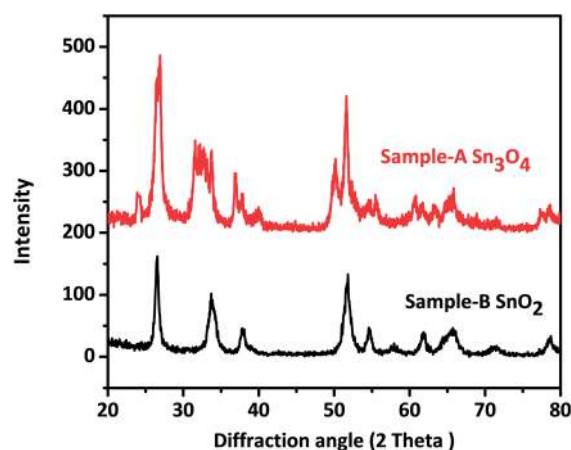


Fig. 1 The X-ray diffraction patterns (XRD) of Sn_3O_4 and SnO_2 .

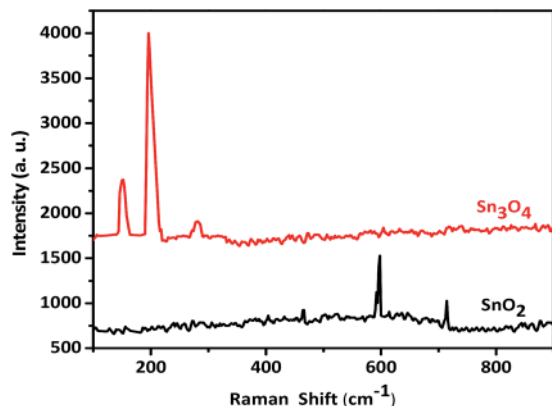


Fig. 2 Room temperature Raman spectra of the as-prepared Sn_3O_4 and SnO_2 .

to main characteristic peaks in triclinic Sn_3O_4 ²³ and, in the case of SnO_2 , the characteristic peaks were observed at 470 and 770 cm^{-1} .²⁴

In the case of the Sn_3O_4 sample, appearance of a weak peak at 245 cm^{-1} may be due to oxidation of the sample.²⁵ This clearly indicates the high purity and crystallinity of the as-prepared sample.

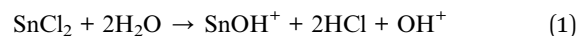
The morphology and microstructural aspects of the samples were examined using FESEM and FETEM. Fig. 3a & b displays the FESEM image of the as-prepared Sn_3O_4 .

It can be seen that the product was mainly composed of discrete nanoplates with layered structures. The layers of nanoplates are clearly seen in the FESEM images. These nanoplates are uneven in size ranging from 100–150 nm. The FESEM images of the SnO_2 sample are shown in Fig. 3c & d. The low magnification image of SnO_2 reveals the formation of microspheres composed of tiny nanoparticles. There may be formation of smaller nanoparticles initially, which may have been subjected to agglomeration under the hydrothermal conditions to form microspheres of 500–700 nm (Fig. 3b).

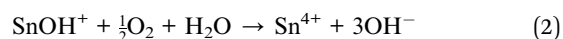
To substantiate the morphological study of the Sn_3O_4 samples, FETEM was performed. Fig. 4a & b clearly demonstrate

the square/rectangular layered plate-like structures of Sn_3O_4 with widths of 100–150 nm and thickness of about 20 nm. Fig. 4c displays the well-resolved lattice fringes (insets of Fig. 3c) with spacing of 0.33 nm and 0.36 nm, which correspond to the (111) and (110) lattice planes of Sn_3O_4 , respectively. This matches well with the XRD results. The corresponding SAED pattern shown in the inset of Fig. 4d depicts the good polycrystallinity of Sn_3O_4 .

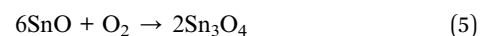
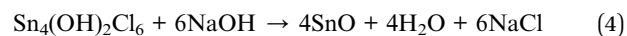
Based on above characterization features, a formation mechanism has been proposed. In our synthetic architecture at pH = 5, pristine SnO_2 was obtained, whereas at pH = 13, pure Sn_3O_4 crystals were obtained. The formation of SnO_2 at pH = 5 can be explained as follows: at acidic pH (pH = 5), the dissolution of Sn(II) species was accompanied with the hydrolysis of SnCl_2 as per eqn (1).



The soluble Sn(II) was further oxidized to insoluble Sn(IV) by the oxygen dissolved in the precursor solution as expressed by eqn (2).



On the other hand, Sn_3O_4 platelets were formed at highly alkaline pH (pH = 13). The formation of Sn_3O_4 at alkaline pH can be explained on the basis of eqn (3)–(5):¹⁹



Thus, the formation of Sn_3O_4 can be attributed to the extent of supersaturation of Sn(II) species under the different

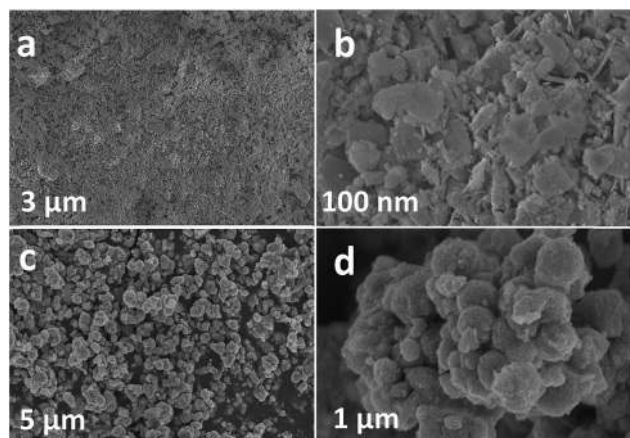


Fig. 3 (a & b) Higher and lower magnification FESEM images of Sn_3O_4 , and (c & d) higher and lower magnification FESEM images of SnO_2 .

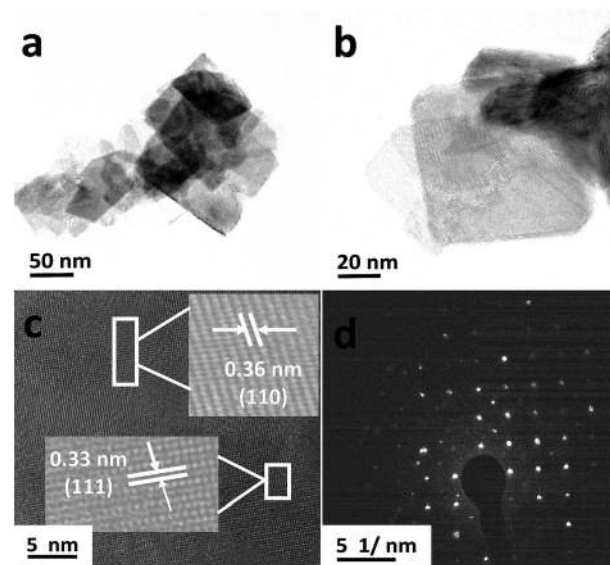
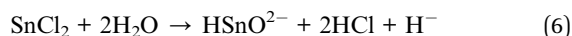


Fig. 4 (a, b & c) TEM images and (d) SAED pattern of Sn_3O_4 .

conditions. To further understand, experiments for the formation of Sn_3O_4 were carried out by decreasing the pH of the reaction mixture (pH = 10, 11 and 12). The pertinent XRD results of the products are given in Fig. 5. From Fig. 5, at pH = 10 the product consists of the SnO phase only. The products obtained at pH = 11 & 12 are comprised of mixed crystals of SnO and Sn_3O_4 . At relatively low alkaline pH (pH = 10), the decreased solubility of Sn(II) species leads to supersaturated conditions and thus, favors the formation of SnO crystals as shown in eqn (4) and the XRD result matches with SnO (JCPDS No. 06-0395). At pH = 11 & 12, the formation of mixed crystals of SnO and Sn_3O_4 can be attributed to the decreased pH of the solution due to the dissolution of SnCl_2 as per eqn (6).



The SnO is further oxidized at higher pH *i.e.* pH = 13, under prolonged hydrothermal conditions as per eqn (5). At higher pH, the sodium chloride restricts further reduction and favors oxidation under self vapour pressure hydrothermal conditions.

On the basis of the morphology studies and the above reaction mechanism, we can easily predict the probable growth mechanism of the plate-like Sn_3O_4 nanostructure. The growth of the plate-like nanostructure proceeds *via* nucleation of tiny Sn_3O_4 nanoparticles and their subsequent evolution into a layered plate shaped nanostructure. In the initial stage, the nanoparticles self-align and exhibit one dimensional growth due to the prolonged hydrothermal conditions. The growth is faster at higher pH and hence the formation of layers of plates is accelerated. Therefore, we observe a dense layered plate-like structure of Sn_3O_4 .

Fig. 6 represents the optical absorption spectra of Sn_3O_4 and SnO_2 and their corresponding Tauc plots. The estimated band gaps for Sn_3O_4 and SnO_2 were found to be 2.76 and 3.21 eV, respectively, which are in good agreement with the reported data.^{26,27} In SnO_2 , all the Sn cations are tetravalent (Sn^{4+}), which make the band gap wide. However, in Sn_3O_4 , both the Sn^{2+} and Sn^{4+} cations co-exist, which are responsible for the narrow band gap.¹⁷

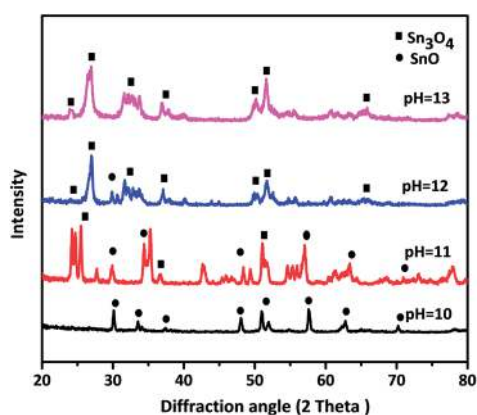


Fig. 5 X-ray diffraction patterns (XRD) of samples at pH = 10, 11, 12 & 13.

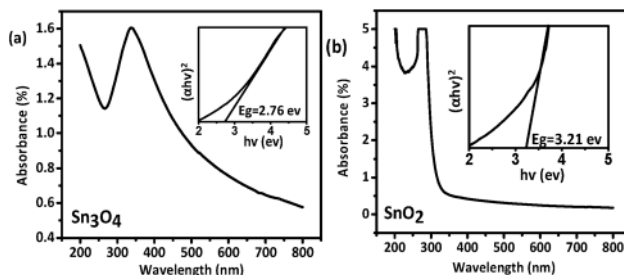


Fig. 6 UV-vis absorption spectra of (a) Sn_3O_4 and (b) SnO_2 . (Inset: the corresponding Tauc plots.)

Generally, the photoluminescence (PL) spectra can provide information on the separation and recombination of photo-induced charge carriers. Fig. 7 shows the PL spectra of Sn_3O_4 and SnO_2 . The intensity of the PL spectrum is directly proportional to the electron-hole recombination rate, *i.e.* the higher the intensity of the spectrum, the higher the rate of electron-hole recombination. In contrast, a lower intensity indicates that more excited electrons are trapped and transferred stably through the interface. The strong emission peaks observed at 334 nm and 363 nm indicate that defects exist in SnO_2 . However, in the case of Sn_3O_4 , the relative peak intensity was decreased. It is well known that strong PL means a high recombination of the photo-generated carriers suggesting low photocatalytic activity. The decrease in the intensity of Sn_3O_4 indicates a minimization of the lattice defects when compared to SnO_2 .

Methylene blue (MB) is a very popular model dye used for transition metal oxide mediated photocatalytic degradation studies. In the current work, the catalytic activities of the as-prepared samples were evaluated by carrying out the dark adsorption and sunlight induced photocatalytic degradation of MB. A control experiment without any catalyst was also performed under sunlight irradiation. P25 (TiO_2) has been studied as a photocatalyst by many research groups due to its merits of inexhaustible abundance with no photo-corrosion and non-toxicity.^{28–30}

In the present study, a direct comparison of the as-prepared samples with P25 was also carried. The changes in the

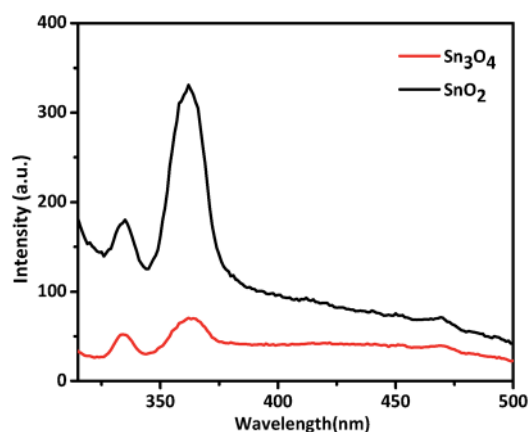


Fig. 7 PL spectra of Sn_3O_4 and SnO_2 .

absorbance spectra of MB dye in aqueous solutions at different time intervals under sunlight illumination were monitored in order to evaluate the photocatalytic activity. Fig. 8 shows the changes in the absorption spectra of an aqueous solution of MB (100 mL, 10 ppm) in the presence of 50 mg of Sn_3O_4 sample under sunlight for different time intervals.

Fig. 8b shows the absorbance of MB dye adsorbed and degraded at different time intervals under the different experimental conditions. The experimental results reveal that MB dye does not degrade under the control experimental conditions. Similarly the reactions carried out in the dark also show no degradation although very little adsorption is seen.

The different samples under sunlight irradiation degraded the MB dye to different extents. The Sn_3O_4 sample demonstrates the largest amount of degradation under sunlight irradiation. The MB concentration was observed to be reduced to 50% within 20 min and completely disappeared after irradiation for 60 min. The SnO_2 sample degraded the MB dye to 80% within 120 min whereas P25 was able to degrade the same amount of MB dye within 70 min.

Thus, the Sn_3O_4 sample exhibits the highest photocatalytic activity among all of the studied photocatalysts. This can be attributed to the band gap of the sample. The reaction kinetics were also determined for the degradation of MB. Fig. 8c illustrates the pseudo first-order kinetic plots of $\ln C_0/C_t$ vs. time for MB degradation using all of the studied samples. The results reveal an excellent linear correlation ($R^2 = 0.974$) for sample-A, suggesting that the degradation follows pseudo first order kinetics with the rate constant $k = 0.046 \text{ min}^{-1}$. The estimated first order rate constants for P25 and SnO_2 were 0.026 and 0.014 min^{-1} , respectively. The data is summarized in Table 1. The high photo-degradation activity of the as-synthesized Sn_3O_4 can be attributed to the nanoplates-like morphology and its narrow band gap of 2.76 eV, which is capable of absorbing visible as well as UV light.

Considering the good spectral response to solar light, the photocatalytic activities of the Sn_3O_4 and SnO_2 nanoparticles

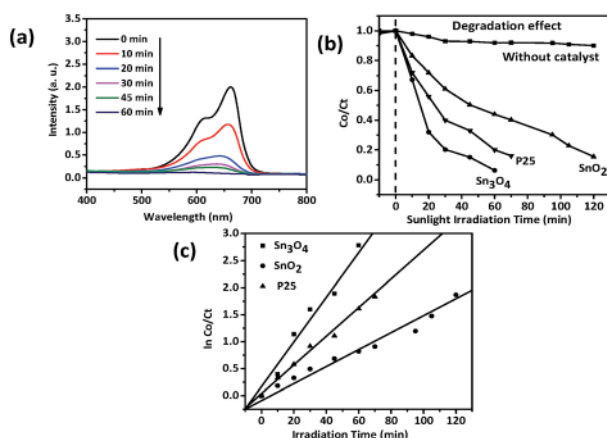


Fig. 8 (a) Spectral changes during the degradation of MB dye in the presence of Sn_3O_4 ; (b) a plot of the change in absorbance vs. irradiation time in the presence of the different catalysts; and (c) the pseudo-first order kinetic plots of $\ln(C_0/C_t)$ vs. irradiation time for the degradation of MB using the different photocatalysts.

Table 1 A summary of the pseudo-first-order kinetics observed for the photocatalytic degradation of methylene blue (MB) using the different photocatalysts

Catalyst	Fitted equation	R^2	$k \text{ (min}^{-1}\text{)}$
Sn_3O_4	$y = 0.046x$	0.974	0.046
SnO_2	$y = 0.014x$	0.978	0.0145
P25 (TiO_2)	$y = 0.026x$	0.989	0.026

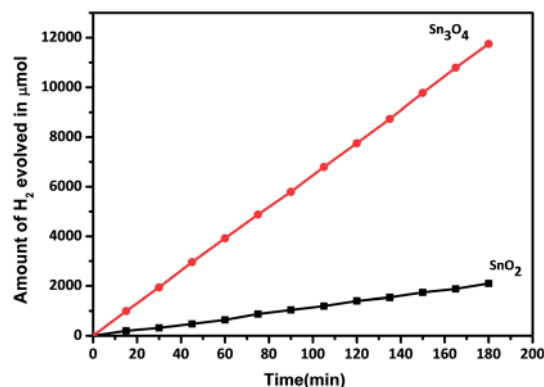


Fig. 9 Photocatalytic hydrogen generation using the Sn_3O_4 and SnO_2 samples.

for hydrogen evolution from H_2S under solar light irradiation have been investigated. Fig. 9 represents the graph of hydrogen evolution with respect to time under solar light. The data for H_2 evolution via H_2S splitting is summarized in Table 2.

The maximum hydrogen evolution ($3916 \mu\text{mol h}^{-1}/0.5 \text{ g}$) was obtained using the Sn_3O_4 nanoparticles, which was much higher than that found for SnO_2 ($729 \mu\text{mol h}^{-1}/0.5 \text{ g}$). However, heterovalent tin oxide clearly exhibited the highest response for hydrogen evolution when compared to univalent tin oxide. It also displayed good stability under the same experimental conditions. The efficiency of the Sn_3O_4 photocatalyst was confirmed by the experiment, which disclosed that in the absence of a photocatalyst as well as under dark conditions, the evolution of hydrogen gas was not observed from a 0.5 M KOH solution flushed with H_2S gas under the visible solar light irradiation. This can also be confirmed from the PL spectra the efficient charge separation and inhibited electron-hole recombination by Sn^{2+} and Sn^{4+} are favorable for enhancing the photocatalytic activity of Sn_3O_4 under sunlight irradiation.

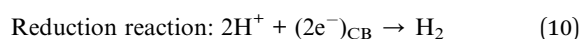
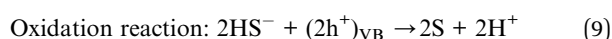
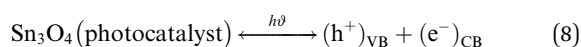
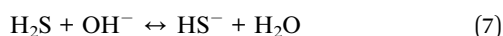
Table 2 A summary of the hydrogen evolution data

Catalyst	Hydrogen evolution rate ($\mu\text{mol h}^{-1}/0.5 \text{ g}$)
Sn_3O_4	3916
SnO_2	729

The stability of the Sn_3O_4 photocatalyst was also studied by repeating the experiments under similar conditions using the same photocatalyst sample after recycling (Fig. 10).

The mechanism of H_2S splitting under solar light irradiation is given by following reactions.³¹

In photocatalytic H_2S splitting, Sn_3O_4 absorbs light energy and generates holes and electrons (h^+ and e^-) pairs (Fig. 11). H_2S , which is weak diprotic acid dissociates at pH 12–13 and maintains equilibrium with hydrosulphide HS^- ions. The photo-induced electrons get transferred to the conduction band (CB) of Sn_3O_4 and provides a reduction site for H^+ to generate H_2 . While the holes in the valence band (VB) of Sn_3O_4 oxidize the HS^- ions into elemental sulphur.



The amount of H_2 evolution from SnO_2 as well as Sn_3O_4 (Table 2) was higher when compared to previous reports.^{17,32} The energy required for H_2S splitting ($\Delta G^\circ = 0.17 \text{ eV}$ or $33.44 \text{ kJ mol}^{-1}$)

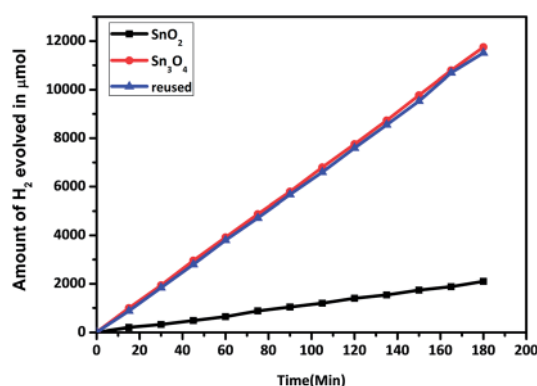


Fig. 10 The reusability study conducted for Sn_3O_4 during the hydrogen generation under sunlight irradiation.

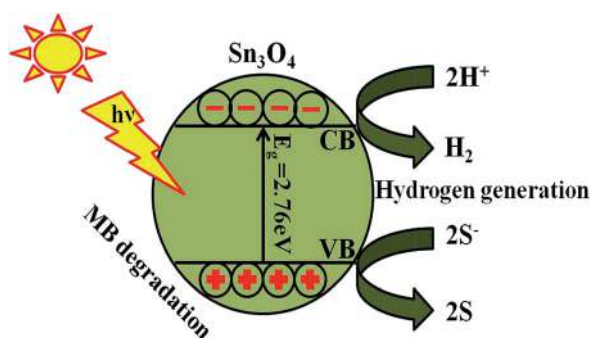


Fig. 11 A schematic representation of the photocatalytic mechanism of Sn_3O_4 .

is much lower than that of water splitting ($\Delta G^\circ = 1.23 \text{ eV}$ or $237.19 \text{ kJ mol}^{-1}$) hence, we observed an enhanced H_2 production.³¹ The conversion of hazardous H_2S gas into eco-friendly fuel (H_2) is an added advantage of this process.

As understood from the optical measurements, the band gap of Sn_3O_4 lies in the visible light region (2.76 eV) and that of SnO_2 in the UV light region (3.21 eV). As discussed in the optical properties section, the narrowing of the band gap of Sn_3O_4 was responsible for the higher activity. This crafts the band structure of Sn_3O_4 better for optical absorption under solar light as well as charge carrier separation. As a result Sn_3O_4 shows enhanced H_2 production over SnO_2 . Further, the PL studies also show the significant inhibition of charge carrier recombination, which was responsible for the higher photocatalytic activity.

4 Conclusions

To conclude, Sn_3O_4 and SnO_2 nanostructures have been synthesized *via* a facile hydrothermal method. pH plays an important role in governing the phase of the synthesized tin oxides. Sn_3O_4 was obtained at pH 13 while at pH 5 the obtained product was SnO_2 . The sunlight driven photocatalytic activity investigations demonstrate that Sn_3O_4 exhibits higher photocatalytic activity towards the degradation of MB as well as hydrogen evolution, even better than P25 and SnO_2 . The superior photocatalytic activity of the as-synthesized Sn_3O_4 sample can be attributed to its photo-absorption ability over a wide spectral range due to the narrowing band gap as well as layered nanoplates-like morphology.

Acknowledgements

Dr P. V. A. and B. B. K. are grateful to ISRO for financial support. Dr D. P. A. is grateful to Prof. Taesung Kim of Sungkyunkwan University and Ministry of Science, ICT and Planning (MSIP, S. Korea) for financial support through the Brain Pool Program of KOFST.

References

- 1 A. Houas, H. Lachheb, M. Ksibi, E. Elaloui, C. Guillard and J. Herrmann, *Appl. Catal., B*, 2001, **31**, 145–157.
- 2 D. F. Ollis, *Environ. Sci. Technol.*, 1985, **19**, 480–484.
- 3 X. B. Chen and S. S. Mao, *Chem. Rev.*, 2007, **107**, 2891–2959.
- 4 J. Li, W. Ma, Y. Huang, X. Tao, J. Zhao and Y. Xu, *Appl. Catal., B*, 2004, **48**, 17–24.
- 5 B. Cheng, J. M. Russell, W. S. Shi, L. Zhang and E. T. Samulski, *J. Am. Chem. Soc.*, 2004, **126**, 5972–5973.
- 6 K. Sakaushi, Y. Oaki, H. Uchiyama, E. Hosono, H. S. Zhou and H. I. mai, *Small*, 2010, **6**, 776–781.
- 7 Y. J. Kim, K. H. Kim, P. Kang, H. J. Kim, Y. S. Choi and W. I. Lee, *Langmuir*, 2012, **28**, 10620–10626.
- 8 M. A. Maki-Jaskari and T. T. Rantala, *Modell. Simul. Mater. Sci. Eng.*, 2004, **12**, 33–41.
- 9 A. Seko, A. Togo, F. Oba and I. Tanaka, *Phys. Rev. Lett.*, 2008, **100**, 45702–45704.
- 10 F. Lawson, *Nature*, 1967, **215**, 955–956.

- 11 M. S. Moreno, R. F. Egerton and P. A. Midgley, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2004, **69**, 233304.
- 12 M. Batzill and U. Diebold, *Prog. Surf. Sci.*, 2005, **79**, 147–154.
- 13 Z. L. Wang, *Annu. Rev. Phys. Chem.*, 2004, **55**, 159–196.
- 14 O. M. Berengue, R. A. Simon, A. J. Chiquito, C. J. Dalmaschio, E. R. Leite, H. A. Guerreiro and F. E. G. Guimarães, *J. Appl. Phys.*, 2010, **107**, 033717.
- 15 C. J. Damaschio, O. M. Berengue, D. G. Stroppa, R. A. Simon, A. J. Ramirez, W. Herwig Schreiner, A. J. Chiquito and E. R. Leite, *J. Cryst. Growth*, 2010, **312**, 2881–2886.
- 16 Y. He, D. Li, J. Chen, Y. Shao, J. Xian, X. Zheng and P. Wang, *RSC Adv.*, 2014, **4**, 1266.
- 17 M. Manikandan, T. Tanabe, P. Li, S. Ueda, G. V. Ramesh, R. Kodiyath, J. Wang, T. Hara, A. Dakshanamoorthy, S. Ishihara, K. Ariga, J. Ye, N. Umezawa and H. Abe, *ACS Appl. Mater. Interfaces*, 2014, **6**, 3790–3793.
- 18 W. Xu, M. Li, X. Chen, J. Zhao, R. Tan, R. Li, J. Li and W. Song, *Mater. Lett.*, 2014, **120**, 140–142.
- 19 M. Li, R. Tan, R. Li, W. Song and W. Xu, *Ceram. Int.*, 2014, **40**, 11381–11385.
- 20 P. H. Suman, A. A. Felix, H. L. Tuller, J. A. Varela and M. O. Orlandi, *Sens. Actuators, B*, 2015, **208**, 122–127.
- 21 Y. H. Li, D. Z. Chen and J. Shao, *RSC Adv.*, 2014, **4**, 1266–1269.
- 22 F. P. Wang, X. T. Zhou, J. G. Zhou and T. K. Z. F. Ding, *J. Phys. Chem. C*, 2007, **111**, 18839–18843.
- 23 Y. Q. Guo, R. Q. Tan, X. Li, J. H. Zhao, Z. L. Luo and C. Cao, *CrystEngComm*, 2011, **13**, 5677–5680.
- 24 W. Z. Wang, C. K. Xu, G. H. Wang, Y. K. Liu and C. L. Zheng, *J. Appl. Phys.*, 2002, **92**, 2740.
- 25 B. Cheng, J. M. Russell, W. S. Shi, L. Zhang and E. T. Samulski, *J. Am. Chem. Soc.*, 2004, **126**, 5972–5973.
- 26 O. M. Berengue, R. A. Simon, A. J. Chiquito, C. J. Dalmaschio, E. R. Leite, H. A. Guerreiro and F. E. G. Guimarães, *J. Appl. Phys.*, 2010, **107**, 033717.
- 27 R. Din, M. H. Farooq, Q. U. A. Javed and D. F. Khan, *Mater. Res. Bull.*, 2012, **47**, 3902–3907.
- 28 Y. Ding, P. Zhang, Z. Long, Y. Jiang, F. Xu and J. Lei, *J. Sol-Gel Sci. Technol.*, 2008, **46**, 176–179.
- 29 A. B. Bocarsly, J. M. Bolts, P. G. Cummins and M. S. Wrighton, *Appl. Phys. Lett.*, 1977, **31**, 568–570.
- 30 A. Sclafani, M. N. Mozzanega and P. Pichat, *J. Photochem. Photobiol., A*, 1991, **59**, 181–189.
- 31 M. A. Mahadadalkar, S. B. Kale, R. S. Kalubarme, A. P. Bhirud, J. D. Ambekar, S. W. Gosawi, M. V. Kulkarni, C. J. Park and B. B. Kale, *RSC Adv.*, 2016, **6**, 34724–34736.
- 32 G. Chen, S. Ji, Y. Sang, S. Chang, Y. Wang, P. Hao, J. Claverie, H. Liu and G. Yu, *Nanoscale*, 2015, **7**, 3117.